



Swathi Sri R 250701769-2025-CSE ▾

S2**Started on** Friday, 11 September 2026, 4:17 PM**State** Finished**Completed on** Friday, 11 September 2026, 4:26 PM**Time taken** 8 mins 31 secs**Marks** 1.00/1.00**Grade** 4.00 out of 4.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that $A[j] - A[i] = k$, $i \neq j$.

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as $5 - 1 = 4$

So Return 1.

For example:

Input	Result
3 1 3 5 4	1

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2  #include <stdlib.h>
3
4  int main() {
5      int n;
6      if (scanf("%d", &n) != 1) return 0;
7
8      int *arr = (int *)malloc(n * sizeof(int));
9      for (int i = 0; i < n; i++) {
10         if (scanf("%d", &arr[i]) != 1) return 0;
11     }
12
13     int k;
14     if (scanf("%d", &k) != 1) return 0;
15
16     int i = 0, j = 1;
17     while (j < n) {
18         int diff = arr[j] - arr[i];
19         if (i != j && diff == k) {
20             printf("1\n");
21             free(arr);
22             return 0;
23         } else if (diff < k) {
24             j++;
25         } else {
26             i++;
27             if (i == j) {
28                 j++;
29             }
30         }
31     }
32
33     printf("0\n");
34     free(arr);
35     return 0;
36 }
37

```

	Input	Expected	Got	
✓	3 1 3 5 4	1	1	✓
✓	10 1 4 6 8 12 14 15 20 21 25 1	1	1	✓
✓	10 1 2 3 5 11 14 16 24 28 29 0	0	0	✓
✓	10 0 2 3 7 13 14 15 20 24 25 10	1	1	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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